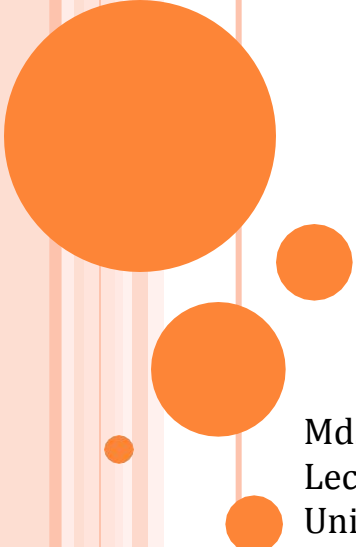


INTRODUCTION TO OOP USING JAVA



Md. Shafqat Talukder Rakin
Lecturer, Department of CSE,
United International University
Email id : shafqat@cse.uiu.ac.bd

Acknowledged: Tanjina Helaly

WHAT IS PROGRAMMING

- Instruction to computer/device to perform task.
- Computer understands only 0 and 1. Nothing else.
- So, we need to send the instruction in the form of 0, 1
 - Do you write program with just 0 and 1?



CLASSIFICATION/EVOLUTION OF PROGRAMMING

- Machine level programming
 - Send instruction in **binary** format
- Assembly Programming
 - send **code** instead of binary code.
 - Need **assembler** to convert to binary
- High level programming
 - Code is **close to English** Language
 - Need **Compiler** to convert to binary
 - 3 types
 - Non structured
 - Structured/Procedural
 - Object Oriented Programming



CLASSIFICATION/EVOLUTION OF PROGRAMMING

❑ Non structured

- Generate spaghetti code
- Sequential and has GoTo
- COBOL, BASIC, FORTRAN

❑ Structured/Procedural

- Use Subroutine/Function
- improving the clarity, quality, and development time
- C, PASCAL

❑ Object Oriented Programming

- Object-oriented programming (OOP) is a programming language model organized around objects rather than "actions" and data rather than logic.
- Historically, a program has been viewed as a logical procedure that takes input data, processes it, and produces output data.
- Java, C++, C#



OUR GOAL

LEARN **OBJECT ORIENTED PROGRAMMING**
USING **JAVA**



PROGRAMMING LANGUAGE

- A programming language is a formal constructed language designed to communicate instructions to a machine, particularly a computer.



JAVA'S LINEAGE

- Java is related to C++, which is a direct descendent of C.
 - Much of the character of Java is inherited from these two languages.
- From C, Java derives its syntax.
- Many of Java's object-oriented features were influenced by C++.



JAVA - CHARACTERISTICS

- Uses C/C++ basic syntax and basic data types -int, char, float, double, long, short, byte etc.
- Uses standard C/C++ control structures
- “Pure” OO language
- No stand alone functions -**All code is part of a class**
- No explicit pointers - uses references
- Uses garbage collection
- Java is strongly typed
- Java is normally compiled to a bytecode.
 - Java bytecode is a machine language for an abstract machine
 - Makes Java secure and Portable
- Each platform (or browser) that runs Java has a Java Virtual Machine (JVM) . The JVM executes Java bytecodes



JAVA – THE PLATFORM

- Java has a large API (application programming interface) covering a wide range of areas The following list of Java APIs and applications from Sun show the range of applications of Java .
 - For reference <http://java.sun.com/products/index.html>
- Java Foundation Classes (JFC) – GUI
- JDBC Database Access
- Java Web Server
- EmbeddedJava - Java on embedded devices



WHY JAVA

- Platform Independent - Code once run anywhere
 - Byte code
- Easy to learn
- Secure
 - Byte code & VM
- Free



JAVA IDE

- Using JDK you can compile and run java program from command line.

- c:> javac HelloWorld. Java

- compiling here and
 - it will produce HelloWorld.class i.e. bytecode.

- c:>java HelloWorld

- It runs java byte code on native machine



JAVA IDE

- Creating, Compiling, Debugging and Execution for these four steps JDK is not user friendly. IDE is provided for that. A list of IDEs are:
 - Eclipse
 - Netbeans.
 - IntelliJ IDEA

You can install IntelliJ IDEA using the tutorial:

[How to Install IntelliJ IDEA on Windows 10 + Creating First Hello World Java Application \(youtube.com\)](#)



AN EXAMPLE HELLOWORLD

```
public class HelloWorldExample
{
    public static void main( String args[] )
    {
        System.out.println("Hello World");
    }
}
```



JAVA SOURCE CODE NAMING CONVENTIONS

- All java source file should end with .java
- Each .java file can contain **only one public class**
- The **name of the file** should be **the name of the public class** plus ".java"
- Do not use abbreviations in the name of the class
- If the class name contains **multiple words** then **capitalize the first letter of each word** ex.
HelloWorld.java



NAMING CONVENTION

▢ *Class Naming*

- *Uses Capitalized word(s) i.e. Title case*
- Examples:- HelloWorld, MyList, StudentMark

▢ *Variable and method names*

- starts with a lowercase letter and after that use Title case
- Examples:- variableAndMethodNames, aFloat, studentName

▢ *Names of constants*

- All are capital letters and separated by underscore.
- Example: NAMES_OF_CONSTANTS



JAVA IDENTIFIERS RULES

- Identifier is a name given to a variable, class, or method.
- Java identifier
 - Can contain letter, number, underscore (_), or dollar sign (\$).
 - Cannot start with number.
 - Identifiers are case sensitive(var and Var both can be declared in a program)
 - have no maximum length.
 - cannot be a keyword, but it can contain a keyword as part of its name.



- Write down whether the following identifiers are valid or not

Name	Valid/invalid	comment
myVar#		
myVar\$		
\$myVar		
final		
static		
finalVar		
1num		
main		



- Write down whether the following identifiers are valid or not

Name	Valid/invalid	comment
myVar#	invalid	# is not allowed
myVar\$	valid	
\$myVar	valid	
final	invalid	keyword
static	invalid	keyword
finalVar	valid	Can contain keyword as a part of name
1num	invalid	Cannot starts with number
main	valid	Not a keyword



Thank You

